Anqi Tan

Biosensing Scientist · Lateral Flow & Point-of-Care Diagnostics

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Ph.D. candidate developing ultrasensitive plasmon-enhanced lateral flow assays and portable readers, validated with human samples in lab and community settings.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Present

Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

RESEARCH EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Developed a competitive plasmonic-fluor lateral flow assay for salivary cortisol with a **≤0.12 ng/mL** detection limit (antibody loading, blocking, and sample-preparation optimization); collected saliva from **153 adults**; accepted for an **oral presentation at APHA 2026**.
- Contribute to a **$4.87M DoD (PRMRP)** project developing a multiplexed IL-8/sTNFR-1 plasmonic-fluor LFA and handheld reader to classify sepsis/ARDS phenotypes at the bedside; lead the reader development.
- Contributed to an ultrasensitive plasmon-enhanced LFA antigen test for *Neisseria gonorrhoeae* with WashU School of Medicine (*ACS Nano Medicine* 2026).
- Built a **~$100 pocket-size fluorescence reader** with phone-controlled capture, comparable to a lab ImageQuant scanner.
- Co-authored 4 journal papers and reviews (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and an Elsevier book chapter on biosensing and point-of-care diagnostics.

Research Assistant Intern, Institute of Brain Science, BrainCo (Hangzhou) May 2022 – Aug 2022

- Completed research tasks for acoustic/electrical-stimulation sleep assistance and multimodal-EEG fatigue monitoring.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

CRISPR Mechanism Explainer (Ashby Prize Hackathon, NCSA) — 1st Place Oct 2025

- Two-person team; found label bias in the public benchmark, built a balanced validation set with experimental-condition context, and fine-tuned an LLM (Qwen3) to classify CRISPR screen outcomes and explain the mechanism; shipped a Gradio chat demo.

HONORS & AWARDS

1st Place, Ashby Prize Computational Science Hackathon (2025) · Honorable Mention, Ashby Prize Hackathon (2024) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021) · 1st Prize, RoboMaster (college level, 2019) · Outstanding Performance & Innovation Award, Intl. Concrete Dragon Boat Competition (2019)

TEACHING & LEADERSHIP

TA, BIOE/ECE 415 Biomedical Instrumentation Lab (Fall 2025) · TA, ME 170 Computer-Aided Design (3 semesters) · President, Residential College Student Committee, ZJU · Volunteer teacher, rural Guizhou (2019)

SKILLS

Biosensing: lateral flow assays (competitive & sandwich), plasmonic-fluor labels, immunoassay optimization, ELISA, saliva/plasma sample prep, 3D cell culture, fluorescence imaging

Engineering: SolidWorks, Creo, Fusion 360, 3D printing, laser cutting, optical/instrument design, COMSOL, FEA, design for assembly

Programming: Python, MATLAB, LabVIEW, Raspberry Pi, Arduino, ESP32

Languages: English (fluent), Chinese (native)